

Nutrition and Aging: Practice Questions (DLA)

1. A 92-year-old woman is brought to the physician because of a 2-week history of burning sensation in her hands and feet and numbness in her lower limbs. She is in a long-term care facility. She has a 6-month history of hypochlorhydria. Laboratory studies show low hemoglobin levels and anemia. Which of the following supplements is most likely to be prescribed to this patient?

- A. Folic acid
- B. Vitamin B12
- C. Magnesium
- D. Vitamin A
- E. Vitamin D

2. A 75-year-old woman comes to the physician for a routine examination. She has recently lost her husband. She has lost 4.5kg (10 lb) since her last visit. She explains that she does not feel the need to cook and her diet consists mainly of canned cream soup and toast. She takes a multivitamin tablet every day. Which of the following conditions is she most likely at risk for?

- A. Protein energy malnutrition
- B. Osteomalacia
- C. Depression
- D. Diverticulosis
- E. Dehydration

3. An 85-year-old man is brought to the physician by his daughter for a routine examination. Laboratory studies show modest elevations of inflammatory signaling proteins and mildly elevated C-reactive protein levels. Which of the following conditions is most likely in this patient?

- A. Protein energy malnutrition
- B. Dysregulation of energy intake
- C. Osteopenia
- D. Chronic inflammatory response
- E. Hypothyroidism

4. A 90-year-old woman is brought to the physician by her caretaker for a routine examination. She has a 30-year history of type-2 diabetes and hypertension. She has maintained good health by keeping fit, taking dietary supplements and taking medication to control her type-2 diabetes and hypertension. Which of the following should most likely be monitored closely during her routine medical visits?

- A. Bone density
- B. White blood cell count
- C. Kidney function
- D. Negative nitrogen balance
- E. Cognitive function

5. A 95-year-old man is admitted to the hospital because he had an accidental fall in his home. He had tripped over his walker while rushing to answer the phone. Which of the following should his family be most concerned about?

- A. Hospital acquired infections
- B. Depression
- C. Loss of muscle mass
- D. Dehydration
- E. All of the above

Answer Key:

- 1. B
- 2. A
- 3. D
- 4. C
- 5. E

Brief explanations:

1. A 92-year-old woman is brought to the physician because of a 2-week history of burning sensation in her hands and feet and numbness in her lower limbs. She is in a long-term care facility. She has a 6-month history of hypochlorhydria (Reduced acid secretion from stomach). Laboratory studies show low hemoglobin levels and anemia. Which of the following supplements is most likely to be prescribed to this patient?

Comment: The patient has hypochlorhydria. HCl is secreted from the parietal cells of stomach. Parietal cell destruction is commonly observed in the elderly. Parietal cells also secrete intrinsic factor which is required for absorption of vitamin B12. Vitamin B12 deficiency results in macrocytic anemia and neurological manifestations (tingling, numbness)

- A. Folic acid – deficiency can also cause macrocytic anemia; but absorption of folate does not require intrinsic factor
- B. **Vitamin B12 – Correct answer – Achlorhydria is a risk factor for vitamin B12 deficiency**
- C. Magnesium – Magnesium deficiency can lead to features similar to calcium deficiency (increased excitability of muscle and nerves)
- D. Vitamin A – Oversupplementation of vitamin A has to be avoided in the elderly
- E. Vitamin D – vitamin D deficiency is common in the elderly due to poor absorption and reduced exposure to sunlight. Vitamin D deficiency can result in osteomalacia characterized by increased tendency to fracture and bone deformities. Serum ALP levels are elevated.

2. A 75-year-old woman comes to the physician for a routine examination. She has recently lost her husband. She has lost 4.5kg (10 lb) since her last visit. She explains that she does not feel the need to cook and her diet consists mainly of canned cream soup and toast. She takes a multivitamin tablet every day. Which of the following conditions is she most likely at risk for?

Comment: Weight loss in the elderly increases muscle protein breakdown which is known as sarcopenia and frailty. Elderly patients are usually in a state of negative nitrogen balance. To conserve muscle mass, it is important to include high quality protein in diet and also to include mobility and exercise.

A. **Protein energy malnutrition – Correct answer – Poor nutrition increases the risk for sarcopenia and frailty in the elderly**

B. Osteomalacia – Vitamin D deficiency is also common in elderly due to reduced absorption and reduced exposure to sunlight. But this patient is supplementing vitamins in her diet.

C. Depression – and Bereavement are risk factors for poor nutrition

D. Diverticulosis

E. Dehydration

3. An 85-year-old man is brought to the physician by his daughter for a routine examination. Laboratory studies show modest elevations of inflammatory signaling proteins and mildly elevated C-reactive protein levels. Which of the following conditions is most likely in this patient?

A. Protein energy malnutrition

B. Dysregulation of energy intake

C. Osteopenia

D. **Chronic inflammatory response – Correct answer – Aging is associated with increased oxidative stress and is generally considered as a chronic inflammatory state**

E. Hypothyroidism

4. A 90-year-old woman is brought to the physician by her caretaker for a routine examination. She has a 30-year history of type-2 diabetes and hypertension. She has maintained good health by keeping fit, taking dietary supplements and taking medication to control her type-2 diabetes and hypertension. Which of the following should most likely be monitored closely during her routine medical visits?

A. Bone density – decreases with age

B. White blood cell count

C. **Kidney function – Correct answer – Renal function in elderly patients taking medications has to be closely monitored. Doses may have to be adjusted based on renal function of the patients.**

D. Negative nitrogen balance – Elderly patients generally have negative nitrogen balance and are frail. Exercise and well-balanced meals can reduce the rate of muscle protein losses

E. Cognitive function

5. A 95-year-old man is admitted to the hospital because he had an accidental fall in his home. He had tripped over his walker while rushing to answer the phone. Which of the following should his family be most concerned about?

A. Hospital acquired infections – Due to poor immunity and immobilization

B. Depression – due to dependency

C. Loss of muscle mass – Sarcopenia or frailty is a common finding in the elderly which can worsen with hospitalization

D. Dehydration

E. **All of the above – Correct answer - People are very fragile at this advanced age, thus he is susceptible to hospital acquired infections. Injury reduces perception of thirst. Muscle wasting happens very fast and he was already dependent on a walker, thus may progress from mildly frail to very frail if physiotherapy is not immediately implemented. Loss of independence can lead to depression.**

Lecture: Synthesis of amino acid derivative hormones

1. A 1-year-old boy is brought to the physician by his parents for a follow up examination. Physical examination and laboratory studies are suggestive of a vitamin deficiency and pyridoxal phosphate deficiency is confirmed. Which of the following reactions would most likely be expected to occur at suboptimal levels in this patient?
 - A. Thyroperoxidase
 - B. Tyrosine hydroxylase
 - C. Phenylethanolamine N-methyl transferase (PNMT)
 - D. Dopa decarboxylase
 - E. Monoamine oxidase
2. A 57-year-old woman comes to the physician for a follow up examination because she has a thyroid gland nodule. As part of her work up, an iodine uptake assay is prescribed to measure activity of her thyroid gland. Which of the following mechanisms is most likely responsible for the uptake of this element into this gland?
 - A. Primary active transport
 - B. Passive diffusion
 - C. Secondary active transport, symport
 - D. Facilitated diffusion
 - E. Secondary active transport, antiport
3. A 52-year-old man comes to the physician because of a 6-month history of panic attacks, palpitations, headache, and sweating. A 24-hour urine analysis shows elevated levels of vanillylmandelic acid. Which of the following is required for the last step in the synthesis of the hormone that is most likely causing the symptoms in this patient?
 - A. Tetrahydrobiopterin
 - B. Pyridoxal phosphate
 - C. Thyroid hormone
 - D. 17 α -hydroxylase (CYP 17)
 - E. S-adenosyl methionine
4. A 37-year-old woman comes to the physician because of a 10-week history of anxiety, hand tremor, heat sensitivity, weight loss, and puffy eyes. Laboratory

show high circulating levels of T3 and T4, and a diagnosis of Graves' disease is made. Which of the following precursors most likely produces the hormone that is most likely responsible for the symptoms in this patient?

- A. Tyrosine
- B. Serotonin
- C. Histidine
- D. Cholesterol
- E. Cholecalciferol

5. A 29-year-old man comes to the physician because of a 1-month history of depression. He says his symptoms worsen during the winter months and seem to ameliorate during the summer. He lives just below the Arctic Circle. Which of the following hormones is most likely associated with the symptoms in this patient?

- A. Dehydroepiandrosterone sulfate (DHEAS)
- B. Thyroid hormone (T3)
- C. Aldosterone
- D. Melatonin
- E. Norepinephrine

Answer Key:

- 1. D
- 2. C
- 3. E
- 4. A
- 5. D

Short Explanations for the answers:

1. A 1-year-old boy is brought to the physician by his parents for a follow up examination. Physical examination and laboratory studies are suggestive of a vitamin deficiency and pyridoxal phosphate deficiency is confirmed. Which of the following reactions would most likely be expected to occur at suboptimal levels in this patient?

- A. Thyroperoxidase – is used for thyroid hormone synthesis in the thyroid gland
- B. Tyrosine hydroxylase – requires tetrahydrobiopterin as a coenzyme

- C. Phenylethanolamine N-methyl transferase (PNMT) – Uses S-adenosyl methionine (SAM) as a methyl group donor
- D. Dopa decarboxylase – Uses pyridoxal phosphate (PLP) to convert DOPA to dopamine (used for amino acid decarboxylation reaction) – CORRECT answer**
- E. Monoamine oxidase – Is used for the inactivation of catecholamines (Epinephrine, norepinephrine and dopamine) and for inactivation of serotonin to 5-HIAA.

2. A 57-year-old woman comes to the physician for a follow up examination because she has a thyroid gland nodule. As part of her work up, an iodine uptake assay is prescribed to measure activity of her thyroid gland. Which of the following mechanisms is most likely responsible for the uptake of this element into this gland?

- A. Primary active transport – Na-K ATPase is a primary active transporter. It uses ATP directly for the transport of these ions. It is an antiport – Sodium moves out of the cell in exchange for K which moves into the cell.
- B. Passive diffusion
- C. Secondary active transport, symport – CORRECT answer – Iodide uptake is by a Sodium-iodide symporter. Sodium moves along its concentration gradient into the cell and pulls iodide with it. Low sodium concentration within the cells are maintained by the Na-K ATPase pump.**
- D. Facilitated diffusion – Glucose transporters (GLUT family) are examples of facilitated diffusion
- E. Secondary active transport, antiport

3. A 52-year-old man comes to the physician because of a 6-month history of panic attacks, palpitations, headache, and sweating. A 24-hour urine analysis shows elevated levels of vanillylmandelic acid. Which of the following is required for the last step in the synthesis of the hormone that is most likely causing the symptoms in this patient?

Comment: The patient has pheochromocytoma. Epinephrine and norepinephrine are inactivated to Vanillyl mandelic acid (VMA) via catechol-O-methyl transferase (COMT) and Monoamine oxidase (MAO)

- A. Tetrahydrobiopterin – Is required as a coenzyme for the conversion of phenylalanine to tyrosine and tyrosine to DOPA (Dihydroxyphenylalanine). The enzymes are phenylalanine hydroxylase and tyrosine hydroxylase. It is also required for tryptophan hydroxylase which forms 5-hydroxy tryptophan from tryptophan for the eventual formation of serotonin (5-hydroxy tryptamine).
- B. Pyridoxal phosphate – is required for DOPA decarboxylase to form dopamine from DOPA. It is also required for 5-hydroxy tryptophan decarboxylase to form serotonin.
- C. Thyroid hormone – synthesis requires iodide and thyroperoxidase.

- D. 17 α -hydroxylase (CYP 17) – Is an enzyme required for the synthesis of steroid hormones
- E. **S-adenosyl methionine – CORRECT answer – PNMT uses SAM as a methyl group donor**

4. A 37-year-old woman comes to the physician because of a 10-week history of anxiety, hand tremor, heat sensitivity, weight loss, and puffy eyes. Laboratory show high circulating levels of T3 and T4, and a diagnosis of Graves' disease is made. Which of the following precursors most likely produces the hormone that is most likely responsible for the symptoms in this patient?

- A. **Tyrosine – CORRECT answer – Tyrosine residues in the thyroglobulin are iodinated to finally result in the synthesis of T4 (thyroxine) and T3. Tyrosine is also a precursor of the catecholamines (epinephrine, norepinephrine and dopamine)**
- B. Serotonin (5-hydroxy tryptamine) – Is formed from tryptophan. Serotonin is a precursor of melatonin (pineal gland)
- C. Histidine – Is a precursor of histamine
- D. Cholesterol – is a common precursor for cortisol, estrogen, testosterone and aldosterone
- E. Cholecalciferol (vitamin D) – is formed from 7-dehydrocholesterol by the action of UV light on skin.

5. A 29-year-old man comes to the physician because of a 1-month history of depression. He says his symptoms worsen during the winter months and seem to ameliorate during the summer. He lives just below the Arctic Circle. Which of the following hormones is most likely associated with the symptoms in this patient?

- A. Dehydroepiandrosterone sulfate (DHEAS) – Is an adrenal androgen and is formed in excess in the congenital adrenal hyperplasias
- B. Thyroid hormone (T3) – Is required for maintenance of BMR and for development of the brain in infants and newborn
- C. Aldosterone – Is a mineralocorticoid. It facilitates the reabsorption of sodium. It also increases potassium excretion and activates H⁺ secretion in the renal tubule. Excessive secretion of aldosterone causes hypertension, hypokalemia and metabolic alkalosis
- D. **Melatonin – secreted from the pineal gland is involved in the sleep-wake cycle and circadian rhythm. It is formed from serotonin which in turn is synthesized from tryptophan.**
- E. Norepinephrine – is a catecholamine. It is formed from Dopamine by dopamine beta-hydroxylase.

Steroid hormone metabolism with explanations

1. A research group investigates de-novo steroid hormone synthesis and uses cholesterol that is only labeled at carbon # 21. Which of the synthesized steroid hormones would most likely contain this label?
 - A. Progesterone
 - B. DHEA
 - C. Androstenedione
 - D. Testosterone
 - E. Estradiol
2. A research group investigates the regulation of de-novo steroid hormone synthesis in different tissues. Which of the following effects is most likely observed when ACTH binds to its receptors?
 - A. Inhibition of cholesteryl esterase
 - B. Activation of StAR
 - C. Stimulation of estradiol synthesis
 - D. Inhibition of P450_{scc}
 - E. Increased release of dihydrotestosterone
3. A clinical study compares the features of patients with congenital adrenal hyperplasia (CAH) resulting from different enzyme deficiencies. All patients have low plasma glucose levels. The adrenal cortex is also highly stimulated. Which of the following steroid hormones is most likely deficient in the observed patients?
 - A. Progesterone
 - B. Aldosterone
 - C. DHEA
 - D. Cortisol
 - E. Androstenedione
4. A 34-year-old woman comes to the physician with a lump in her right breast. Biopsy studies show a specific type of breast cancer. She is prescribed a drug that acts as an aromatase (CYP19) inhibitor. Synthesis of which of the following hormones would most likely be reduced following administration of this inhibitor?
 - A. Cortisol
 - B. Progesterone
 - C. Estradiol
 - D. DHEA
 - E. Aldosterone
5. Thirty women aged 20- 30 years volunteer for a study on hormonal and metabolic changes following stress. They are all exposed to a stressor, and it is observed that

their anterior pituitary glands secrete a specific proteohormone. The secretion is feedback inhibited by a specific steroid hormone formed in the z. fasciculata of the adrenal gland. Which of the following best describes the steroid hormone released from the adrenal gland in these women?

- A. Overproduction leads to Addison disease
- B. Stimulates skeletal muscle protein synthesis
- C. Steroid hormone with 18 carbons
- D. Deficiency results in Cushing Syndrome
- E. Reduces inflammation and the synthesis of all eicosanoids

6. A researcher is studying patients with congenital adrenal hyperplasia (CAH) which is a group of rare genetic diseases that change the pattern of steroid hormone synthesis in the adrenal cortex. They analyze the clinical and laboratory findings in patients confirmed to have an inherited deficiency of CYP17 enzyme. Which of the following is most likely observed in these patients?

- A. Aldosterone deficiency
- B. Cortisol deficiency
- C. Increased DHEA synthesis
- D. Increased pregnenolone release
- E. Ambiguous external genitalia in female infants

7. A 28-year-old man comes to the physician for a follow up examination. He has been on long-term treatment with high dose corticosteroids. Physical examination shows changes in body fat distribution similar to Cushing Syndrome. Laboratory studies show elevated plasma glucose level. Which of the following is most likely responsible for this laboratory finding?

- A. Increased gluconeogenesis in hepatocytes
- B. Increased triacylglycerol (TAG) degradation in fat cells
- C. Inhibited muscle protein synthesis
- D. Inhibited release of ACTH from the anterior pituitary gland
- E. Inhibited release of corticotropin-releasing hormone (CRH) from hypothalamus

8. A 24-year-old man comes to the physician because of a 1-month history of weakness and tiredness. Clinical and laboratory studies confirm a diagnosis of tuberculosis of the adrenal gland that has resulted in destruction and atrophy of the adrenal cortex. Which of the following findings is most likely observed in this patient?

- A. Increased upper body fat
- B. High blood pressure
- C. High serum sodium levels
- D. Bronze pigmentation of skin
- E. Low levels of circulating ACTH

9. A new-born girl is born at full-term. Physical examination shows ambiguous male-like genitalia. Her blood pressure is low. Laboratory studies show low fasting plasma

glucose, hyponatremia and hyperkalemia. Which of the following enzymes is most likely deficient in this patient?

- A. CYP11A1 (desmolase)
- B. CYP 17 (17-hydroxylase).
- C. CYP 21 (21-hydroxylase).
- D. CYP 11B1 (11 β -hydroxylase).
- E. Aldosterone synthase.

10. A new-born girl is born at full-term. Physical examination shows ambiguous male-like genitalia. Her blood pressure is mildly elevated. Laboratory studies show low plasma glucose level. Further laboratory studies show elevated levels of 11-deoxycorticosterone but low corticosterone and normal 17-hydroxy-progesterone levels. Which of the following enzymes is most likely deficient in this patient?

- A. 3-hydroxysteroid dehydrogenase.
- B. CYP 17 (17 hydroxylase).
- C. CYP 21 (21 hydroxylase).
- D. CYP 11B1 (11 β -hydroxylase).
- E. CYP 11 A1 (desmolase).

Answer key: 1 A, 2 B, 3 D, 4 C, 5 E, 6 B, 7 A, 8 D, 9 C and 10 D.

SHORT EXPLANATIONS AND COMMENTS

1. A research group investigates de-novo steroid hormone synthesis and uses cholesterol that is only labeled at **carbon # 21**. Which of the synthesized steroid hormones would most likely contain this label?

Comment: easy Q when you know that cholesterol has 27 carbons and that P450_{scc} cleaves the side chain to pregnenolone with 21 C. Progesterone, aldosterone and cortisol all have 21 carbons and would have the label. Only progesterone is a choice in this question.

- A. **Progesterone** Yes correct: Progesterone is formed by 3 hydroxysteroid DH from pregnenolone.
- B. DHEA no it has 19 C and the label would not be found
- C. Androstenedione no, it has only 19 carbons
- D. Testosterone no, it has only 19 carbons
- E. Estradiol no, it has only 18 C

2. A research group investigates the regulation of de-novo steroid hormone synthesis in different tissues. Which of the following effects is most likely observed when **ACTH** binds to its **receptors**?

Comment: First you have to know that ACTH receptors are found in the adrenal cortex and not in testes or ovaries. You have to know that ACTH acts via protein kinase-A (PKA) and activates cholesteryl esterase and StAR

- A. Inhibition of cholesteryl esterase **no**, this enzyme is activated. Cholesteryl esters are stored and only the free cholesterol can be transported into mitochondria and is a substrate for P450_{scc} (CYP 11A)
- B. **Activation of StAR** Yes, ACTH uses the cAMP messenger system and activates StAR which transports free cholesterol into mitochondria.
- C. Stimulated synthesis of estradiol **no**, ACTH binds to the adrenal cortex receptors. Estradiol is mostly synthesized in ovaries or fat cells from androgens and activation of StAR is by far the best answer for this question.
- D. Inhibition of P450_{scc} **no**, the cholesterol side-chain cleavage enzyme is mainly regulated by substrate availability in mitochondria. Only as a follow-up of cholesterol transport into mitochondria by StAR, the enzyme is activated and definitely not inhibited.
- E. Stimulated release of dihydrotestosterone **no**, ACTH acts on the adrenal cortex and leads to release of weak androgens. Testosterone is released from testes by Leydig cells and dihydrotestosterone is formed by 5-alpha-reductase in the prostate and other cells.

3. A clinical study compares the features of patients with congenital adrenal hyperplasia (CAH) resulting from different enzyme deficiencies. All patients have **low plasma glucose levels**. The adrenal cortex is also highly stimulated. Which of the following steroid hormones is most likely deficient in the observed patients?

Comment: CAH results from an enzyme deficiency in cortisol synthesis. This can be 3-hydroxysteroid DH (rare), CYP 17, CYP 21 or CYP 11B. Cortisol is the most correct answer. All patients have reduced amounts of cortisol which leads to the overstimulation of the adrenal cortex by ACTH. In CAH, low cortisol levels does not provide the normal feedback inhibition of ACTH release and hence ACTH levels would be highly elevated. Low cortisol often leads to low plasma glucose levels (hypoglycemia).

- A. Progesterone **no**, 3-hydroxysteroid DH is a rare disease and low levels of progesterone would not be found in all patients, who mostly have CYP 21 (most common) or CYP 17 or CYP 11 B deficiency.
- B. Aldosterone **no**, it is only deficient in CAH patients with 3-hydroxysteroid DH, CYP 21 or CYP 11B deficiencies.
- C. DHEA **no**, it is only deficient in CAH due to CYP 17 deficiency

- D. **Cortisol** **Yes**, it is synthesized in reduced amounts in all CAH patients and this leads to overstimulation of the adrenal cortex by ACTH and often to low blood glucose levels
- E. Androstenedione **no**, it is only deficient in CAH due to 3-hydroxysteroid DH deficiency or CYP 17 deficiency.

4. A 34-year-old woman comes to the physician with a lump in her right breast. Biopsy studies show a specific type of **breast cancer**. She is prescribed a drug that acts as an **aromatase (CYP19) inhibitor**. Synthesis of which of the following hormones would most likely be reduced following administration of this inhibitor?

Comment: easy question when you know that only estrogens contain an aromatic ring A in comparison to other steroid hormones. Note, estrogen-responsive breast cancer is often treated with tamoxifen (interference with estrogen receptor binding) or with aromatase inhibitor drugs (reduction of estradiol synthesis).

- A. Cortisol **no**, aromatase is not used
- B. Progesterone **no**, aromatase is not used
- C. **Estradiol** **Yes**, aromatase (CYP 19) forms the aromatic ring. Estrogens are synthesized from androgens. Testosterone (C19) using aromatase leads to estradiol (C18, aromatic ring A which is needed for estrogen receptor recognition).
- D. DHEA **no**, aromatase is not used
- E. Aldosterone **no**, aromatase is not used

5. Thirty women aged 20- 30 years volunteer for a study on hormonal and metabolic changes following **stress**. They are all exposed to a stressor and it is observed that their **anterior pituitary glands** secrete a specific **proteohormone**. The secretion is feedback inhibited by a specific **steroid hormone** formed in the **z. fasciculata** of the adrenal gland. Which of the following best describes the steroid hormone released from the adrenal gland in these women?

Comment: short stem with many layers, more complex question but doable!

First you have to know that stress stimulates ACTH release, then you have to know that cortisol inhibits ACTH release. As help, it is mentioned that the steroid hormone is formed in the z. fasciculata, this makes it easy as you know that it is cortisol. Finally you have to know the characteristics for cortisol.

- A. Overproduction leads to Addison disease **no**, Addison disease results from **less production of cortisol in the adrenal cortex due to atrophy**.
- B. Stimulates skeletal muscle protein synthesis **no**, cortisol stimulates muscle protein **degradation**

- C. Steroid hormone with 18 carbons **no**, cortisol has **21** carbons, 18 C is found for **estrogens**
- D. Deficiency results in Cushing Syndrome **no**, cortisol **overproduction** due to a tumor of the **z. fasciculata** is found in Cushing Syndrome or overproduction of CRH due to tumor in hypothalamus (then named Cushing disease)
- E. **Reduces inflammation and synthesis of eicosanoids** Yes, correct. Elevated cortisol levels inhibits phospholipase A2 during an inflammation. Phospholipase A2 releases arachidonic acid needed for eicosanoid synthesis from membrane phospholipids. When free arachidonic acid is not formed, then eicosanoid synthesis is reduced. Eicosanoids are the group of prostaglandins which include, PGE2 (inflammatory mediator), PGI₂ (prostacyclin), thromboxane and the leukotrienes.

6. A researcher is studying patients with congenital adrenal hyperplasia (CAH) which is a group of rare genetic diseases that change the pattern of steroid hormone synthesis in the **adrenal cortex**. They analyze the clinical and laboratory findings in patients confirmed to have an inherited deficiency of **CYP17 enzyme**. Which of the following is most likely observed in these patients?

- A. Aldosterone deficiency
- B. Cortisol deficiency
- C. Increased DHEA synthesis
- D. Increased pregnenolone release
- E. Ambiguous external genitalia in female infants

Comment: Features of CAH are always asked and you know that they all have in common that one enzyme of cortisol synthesis is deficient. CYP 17 is an enzyme needed for cortisol synthesis and this makes this question really easy.

- A. Aldosterone deficiency **no**, aldosterone is **increased** in CYP 17 deficiency as it's synthesis does not need this enzyme and the normal regulations are overruled by continuous ACTH overstimulation. Increased aldosterone synthesis results in **hypertension**.
- B. **Cortisol deficiency** Yes, correct, as CYP 17 is needed for cortisol synthesis. All CAH show deficiency of cortisol and often low blood glucose levels.
- C. Increased DHEA synthesis **no**, CYP 17 is needed for DHEA synthesis. Aldosterone synthesis is stimulated.
- D. Increased pregnenolone release **no**, pregnenolone is formed by P450_{scc} but it is not released from the adrenal cortex. It is used to form progesterone and leads to more aldosterone in CYP 17 deficiency.
- E. Ambiguous external genitalia in female infants **No**, CYP17 is needed for androgen synthesis. When it is deficient, less androgens are formed and female outer genitalia look normal. Abnormal outer genitals are found for male infants (ambiguous, female-like).

7. A 28-year-old man comes to the physician for a follow up examination. He has been on **long-term treatment** with **high dose corticosteroids**. Physical examination shows changes in body fat distribution similar to Cushing Syndrome. Laboratory studies show **elevated plasma glucose level**. Which of the following is most likely responsible for this laboratory finding?

Comment: You have to know that corticosteroid drugs are similar to cortisol. It is helpful that **Cushing Syndrome** is mentioned. It is also important to notice that it is asked what **causes his hyperglycemia**, and the **best correct answer** is **stimulated gluconeogenesis** in hepatocytes.

- A. **Increased gluconeogenesis in hepatocytes** Yes, correct. Under normal conditions, cortisol is released to prevent hypoglycemia and cortisol stimulates gluconeogenesis during prolonged starvation or stress situations. Treatment with corticosteroids interferes with normal regulations, and hepatic gluconeogenesis is stimulated leading to **hyperglycemia**.
- B. Increased triacylglycerol (TAG) degradation in fat cells no, although it takes place, this is **not the cause for hyperglycemia**. Cortisol stimulates TAG degradation in fat cells and these fatty acids are released into the blood and are mostly used by other cells for energy metabolism. In hepatocytes, fatty acid degradation is needed for energy metabolism and is needed to allow the switch from glycolysis to gluconeogenesis.
- C. Inhibited muscle protein synthesis no, Cortisol leads to muscle protein **degradation**. The amino acids formed by protein degradation are used to form alanine which is then released into the blood and can be used by hepatocytes for gluconeogenesis.
- D. Inhibited release of ACTH from the pituitary gland no, this is **not the cause for hyperglycemia**, although corticosteroid drugs have the action similar to cortisol and inhibit ACTH release
- E. Inhibited release of corticotropin-releasing hormone from the hypothalamus no, this is **not the cause for hyperglycemia**, although corticosteroid drugs have the action similar to cortisol.

8. A 24-year-old man comes to the physician because of a 1-month history of weakness and tiredness. Clinical and laboratory studies confirm a diagnosis of tuberculosis of the adrenal gland that has resulted in **destruction and atrophy of the adrenal cortex**. Which of the following findings is most likely observed in this patient?

Comment: First you have to make the connection to **Addison disease**, then it is an easy question when you know the **clinical signs** of Addison disease. Hypoglycemia, hypotension, low serum sodium levels, hyperkalemia, pigmentation of skin and mucus membranes are clinical features of Addison disease.

- A. Increased upper body fat **no**, this is seen in Cushing syndrome.
- B. High blood pressure **no**, due to his adrenal atrophy, he has less aldosterone synthesis and hypotension.
- C. High serum sodium levels **no**, the lack of aldosterone in Addison disease leads to low sodium levels (hyponatremia) and to high potassium levels (hypernatremia) and salt craving.
- D. **Bronze pigmentation of skin Yes, correct.** Cortisol levels in Addison disease are low and not sufficient for feedback inhibition of CRH release. This leads to more POMC which will be cleaved into smaller hormones, like ACTH but also to melanocyte stimulating hormone. This leads to the bronze pigmentation of the skin and gum line.
- E. Low levels of circulating ACTH **no**, ACTH levels are high as the ACTH release from the pituitary gland (and CRH from the hypothalamus) are not feed-back inhibited by cortisol.

9. A new-born **girl** is born at full-term. Physical examination shows **ambiguous male-like genitalia**. Her **blood pressure is low**. Laboratory studies show **low fasting plasma glucose, hyponatremia and hyperkalemia**. Which of the following enzymes is most likely deficient in this patient?

Comment: Low blood glucose and hypotension, hyponatremia and hyperkalemia indicate CAH with low levels of mineralocorticoids. Her adrenal androgens would be elevated. Routine screening would show elevated 17-OH-progesterone levels.

- A. CYP 11A1 (desmolase) **no** that would not lead to elevated adrenal androgens
- B. CYP 17 (17 hydroxylase) **no** that would lead to low androgens and hypertension
- C. **CYP 21 (21 hydroxylase) Yes, correct. It leads to the clinical signs of the patient. It can lead to a “salt crisis” when untreated.**
- D. CYP 11B1 (11 β -hydroxylase) **no** this would lead to normal or mildly elevated blood pressure.
- E. Aldosterone synthase **no** this would not lead to low blood glucose and CAH

10. A new-born **girl** is born at full-term. Physical examination shows **ambiguous male-like genitalia**. Her **blood pressure is mildly elevated**. Laboratory studies show **low plasma glucose level**. Further laboratory studies show **elevated levels of 11-deoxycorticosterone** but **low corticosterone** and normal 17-hydroxy-progesterone levels. Which of the following enzymes is most likely deficient in this patient?

Comment: She is diagnosed with CAH but CYP21 deficiency is ruled out, not only by the elevated blood pressure but also normal levels of 17-OH-progesterone.

- A. 3-hydroxysteroid dehydrogenase **no** then all mineralocorticoids would be low
- B. CYP 17 (17-hydroxylase) **no** that would lead to low androgens; female children generally have normal external genitalia, but delayed puberty.
- C. CYP 21 (21-hydroxylase) **no** that would lead to accumulation of 17-OH-progesterone and low blood pressure.
- D. **CYP 11B1 (11beta-hydroxylase) Yes, correct, 11-deoxycorticosterone and adrenal androgens accumulate leading to normal to mildly elevated blood pressure and ambiguous male-like genitalia.**
- E. CYP 11 A1 (desmolase) **no** then all corticoids would be low

Endocrine Pancreas, Diabetes mellitus and Hypoglycemia with explanations

Suggestion: take all 10 MCQ in a row in 15 min and then read the explanations.

1. A pharmaceutical company develops a new drug for the treatment of patients with type 2 diabetes mellitus. The drug increases insulin secretion from the beta-cells of the pancreas. They study the intracellular changes which occur following administration of the drug. Which of the following intracellular events is most likely observed following administration of the drug?

- A. Inhibition of opening of voltage-gated calcium channels.
- B. Inhibition of ATP-sensitive potassium channels.
- C. Inhibition of glucokinase.
- D. Low cellular ATP levels
- E. Inhibition of GLUT-2 transporters

2. A research group investigates the factors responsible for synthesis and release of glucagon in the pancreatic alpha-cells. Which of the following most likely activates the release of this hormone?

- A. Glucagon-like peptide-1 (GLP-1)
- B. Sympathetic stimulation
- C. Somatostatin
- D. Hyperglycemia
- E. Leucine

3. A 65-year-old man is admitted to hospital. Laboratory studies show fasting blood glucose is 60 mg/dL (Normal range 70-100 mg/dL) following an overnight fast. Which of the following cell functions is most likely impaired in this patient?

- A. Type 2 alveolar cells
- B. Fat cells
- C. Red blood cells
- D. Hepatocytes
- E. Renal medulla cells

4. A 34-year-old woman comes to the physician because of a 1-month history of worsening generalized skin pigmentation. Clinical and laboratory studies confirm a diagnosis of Addison disease. She is prescribed corticosteroids. Which of the following pathways is most likely activated following this treatment?

- A. Protein synthesis in skeletal muscle cells
- B. Fatty acid synthesis in hepatocytes
- C. Gluconeogenesis in hepatocytes
- D. Glycogen synthesis in skeletal muscle cells
- E. TAG synthesis in fat cells

5. A 24-year-old woman is brought to the emergency department by her friend because of a 2-hour history of dizziness, profuse sweating and palpitations. Laboratory studies show her plasma glucose level is 55 mg/dL (Normal range 70-100 mg/dL). She is given orange juice. She is also prescribed a hormone injection which stimulates hepatic glycogen degradation and rapidly increases her plasma glucose levels. Which of the following hormones most likely results in the observed effect?

- A. Dopamine
- B. Somatostatin
- C. Insulin
- D. Glucagon like peptide-1 (GLP-1)
- E. Epinephrine

6. A 36-year-old woman is brought to the emergency department because of a 1-hour history of dizziness and palpitations. Laboratory studies show severe hypoglycemia which is managed immediately. She has a history of depression. She has previously used her son's insulin injections used for management of his type-1 diabetes mellitus. Her blood samples are obtained for estimation of insulin, proinsulin and C-peptide. Which of the following laboratory findings are most likely in this patient?

- A. High C-peptide, increased proinsulin, low insulin
- B. High C-peptide, absent proinsulin, high insulin
- C. Low C-peptide, increased proinsulin, low insulin
- D. Low C-peptide, absent proinsulin, high insulin
- E. High C-peptide, increased proinsulin, high insulin

7. A group of thirty healthy men aged 25-30 years volunteer for a study that investigates blood glucose levels during prolonged starvation. On day 3 of starvation they have an average fasting blood glucose level of 70 mg/dL (Normal range 70-100 mg/dL). Which of the following blood glucose level is most likely expected on day 10 of starvation in these volunteers?

- A. 100 mg/dL
- B. 90 mg/dL
- C. 70 mg/dL
- D. 55 mg/dL
- E. 40 mg/dL

8. A research group investigates the synthesis and release of insulin in beta-cells of pancreas. Which of the following best describes the function of C-peptide in this process?

- A. Needed for release of insulin granules
- B. Is added to insulin to form proinsulin
- C. Allows optimum formation of all three disulfide bonds
- D. Increases the half-life of insulin in blood
- E. Links the A- and B-chains of insulin in circulation

9. A research group investigates the factors affecting the synthesis and release of insulin in pancreatic beta-cells. Which of the following most likely activates the release of this hormone?

- A. Sympathetic stimulation
- B. Glucagon-like peptide-1 (GLP-1)
- C. Somatostatin
- D. Cortisol
- E. Alanine

10. A research group investigates the action of insulin on human skeletal muscle. The volunteers for this study are adolescents aged 14-17 years. All of them are provided a healthy diet containing all dietary essential amino acids. The metabolic changes in

skeletal muscle following the meal are observed. Which of the following effects would most likely be observed in these volunteers?

- A. Increased synthesis of sodium-potassium ATPase.
- B. Decreased uptake of amino acids and glucose into muscle cells.
- C. Negative nitrogen balance
- D. Activation of GLUT-2 transporters for uptake of glucose.
- E. Decreased muscle protein synthesis

Answer key:

1 B, 2 B, 3 D, 4 C, 5 E, 6 D, 7 C, 8 C, 9 B, 10 A .

Explanations:

1. A pharmaceutical company develops a new drug for the treatment of patients with type 2 diabetes mellitus. The drug **increases** insulin secretion from the beta-cells of the pancreas. They study the intracellular changes which occur following administration of the drug. Which of the following intracellular events is most likely observed following administration of the drug?

- A. Inhibition of opening of voltage-gated calcium channels.
- B. Inhibition of ATP-sensitive potassium channels.
- C. Inhibition of glucokinase.
- D. Low cellular ATP levels
- E. Inhibition of GLUT-2 transporters

Comment: The new drug most likely has a mechanism similar to sulfonylurea. It increases the release of insulin and inhibits the ATP-sensitive potassium channels in the beta-cells of pancreas.

- A. Inhibition of opening of voltage-gated calcium channels **no the influx of calcium into the cells is necessary to trigger the release of insulin granules.**
- B. **Inhibition of ATP-sensitive potassium channels. Yes, correct.** A high level of ATP resulting from high blood glucose leads to high enough ATP levels to inhibit the ATP-sensing potassium channels. This is also the target for sulfonylurea which is now commonly used to treat Diabetes Mellitus Type II.

- C. Inhibition of glucokinase. **no**, this would reduce insulin release. Note: There is a hereditary deficiency of glucokinase and this leads to MODY-2, a special Type of Diabetes Mellitus. Due to less activity of glucokinase, higher levels of blood glucose are needed to form the high ATP level in the cell to inhibit the ATP-sensing potassium channel and to release insulin.
- D. Low cellular ATP levels **no**, on the contrary, **high** ATP level is needed
- E. Inhibition of GLUT-2 transporters **no** GLUT-2 transporters have to function and to transport glucose into cells by facilitated glucose transport along the concentration gradient.

2. A research group investigates the factors responsible for synthesis and release of **glucagon** in the pancreatic **alpha-cells**. Which of the following most likely **activates** the release of this hormone?

Comment: it is important to know the regulations of both, insulin and glucagon release and know the incretins that enhance insulin release.

- A. Glucagon-like peptide 1 (GLP-1) **no** this is an incretin and activates the release of insulin
- B. **Sympathetic stimulation Yes, correct.** Glucagon is released during flight and fight situations and the low blood glucose level is recognized by the hypothalamus.
- C. Somatostatin **no**, this hormone inhibits the release of glucagon and also of insulin and is involved with fine tuning.
- D. Hyperglycemia **no** Hyperglycemia inhibits glucagon secretion and stimulates the release of insulin from beta-cells of pancreas.
- E. Leucine **no** this amino acid stimulates release of insulin. Glucagon release is stimulated by arginine and alanine.

3. A 65-year-old man is admitted to hospital. Laboratory studies show fasting blood glucose is 60 mg/dL (Normal range 70-100 mg/dL) following an overnight fast. Which of the following cell functions is most likely impaired in this patient?

Comment: Blood glucose levels leading to hypoglycemia are often related **to less efficient hepatocytes** or to abnormal **hormone levels**. The choices show cells and not hormones, so hepatocytes are the correct choice.

- A. Type 2 alveolar cells **no** they are important for lung surfactant synthesis but not for gluconeogenesis
- B. Fat cells **no** their function is not directly related to blood glucose levels
- C. Red blood cells **no** damage of these cells results in hemolysis and not hypoglycemia.
- D. **Hepatocytes Yes, correct.** Hepatocytes perform both glycogen degradation and gluconeogenesis during an overnight fast. They may be damaged in this patient.

- E. Renal medulla cells **no** it is the **renal cortex** cells that are involved in gluconeogenesis and release of glucose into the blood. Renal medulla cells are mostly performing glycolysis.

4. A 34-year-old woman comes to the physician because of a 1-month history of worsening generalized skin pigmentation. Clinical and laboratory studies confirm a diagnosis of Addison disease. She is prescribed **corticosteroids**. Which of the following pathways is most likely **activated** following this treatment?

Comment: Addison disease results from destruction of the adrenal glands. This leads to **deficient** release of **all adrenal cortical hormones**. The choices show pathways that are related to cortisol effects and you have to know the metabolic outcome of cortisol actions.

- A. Protein synthesis in skeletal muscle cells **no** on the contrary, protein degradation takes place
- B. Fatty acid synthesis in hepatocytes **no** **insulin** stimulates fatty acid synthesis in hepatocytes
- C. **Gluconeogenesis in hepatocytes** **Yes, correct.** Cortisol induces enzymes of gluconeogenesis in the liver.
- D. Glycogen synthesis in skeletal muscle cells **no** glycogen synthesis is stimulated by **insulin**
- E. TAG synthesis in fat cells **no**, TAG **degradation** in fat cells is stimulated by cortisol

5. A 24-year-old woman is brought to the emergency department by her friend because of a 2-hour history of dizziness, profuse sweating and palpitations. Laboratory studies show her plasma glucose level is 55 mg/dL (Normal range 70-100 mg/dL). She is given orange juice. She is also prescribed a hormone injection which **stimulates hepatic glycogen degradation** and rapidly **increases her plasma glucose levels**. Which of the following hormones most likely results in the observed effect?

Comment: It is important to know the group of **insulin counterregulatory hormones** and to know what they are doing.

- A. Dopamine **no** this hormone (neurotransmitter) is released in the brain and is not directly involved with blood glucose levels
- B. Somatostatin **no** this hormone fine tunes the release of glucagon and insulin but is not grouped as insulin counterregulatory hormone
- C. Insulin **no** this hormone leads to reduction of high blood glucose levels
- D. Glucagon like peptide-1 (GLP-1) **no** This hormone is an incretin and stimulates insulin release
- E. **Epinephrine** **Yes, correct.** This hormone leads to hepatic glycogen degradation.

6. A 36-year-old woman is brought to the emergency department because of a 1-hour history of dizziness and palpitations. Laboratory studies show **severe hypoglycemia** which is managed immediately. She has a history of depression. She has previously used her son's **insulin injections** used for management of his type-1 diabetes mellitus. Her blood samples are obtained for estimation of insulin, proinsulin and C-peptide. Which of the following laboratory findings are most likely in this patient?

Comment: commonly tested are insulin, C-peptide and proinsulin and if needed sulfonylurea.

- A. High C-peptide, increased proinsulin, low insulin **no** High C-peptide and high proinsulin are found in insulinoma; however insulin levels are also elevated
- B. High C-peptide, absent proinsulin, high insulin **no** this combination is mostly not found; C-peptide and proinsulin are elevated in insulinoma
- A. Low C-peptide, increased proinsulin, low insulin **no** this combination is mostly not found
- B. Low C-peptide, absent proinsulin, high insulin **Yes**, this combination indicates insulin injection. C-peptide and proinsulin levels are low due to inhibition of endogenous insulin release because of high exogenous insulin levels in the circulation
- C. High C-peptide, increased proinsulin, high insulin **no** High C-peptide and high proinsulin and elevated insulin levels are found in insulinoma

7. A group of thirty healthy men aged 25-30 years volunteer for a study that investigates blood glucose levels during prolonged starvation. On day 3 of starvation they have an average fasting blood glucose level of 70 mg/dL (Normal range 70-100 mg/dL). Which of the following blood glucose level is most likely expected on day 10 of starvation in these volunteers?

Comment: the blood glucose levels drops after 3 days of starvation to about 70 mg/ml. After that it does not raise and neither does drop lower in healthy individuals.

- A. 100 mg/dL **no**, it does not increase again to normal levels
- B. 90 mg/dL **no**, it does not increase again to normal levels
- C. **70 mg/dL** **Yes, correct**, generally it stays in this range all throughout starvation
- D. 55 mg/dL **no**, this is not expected, it leads to an adrenergic response in hypoglycemic individuals
- E. 40 mg/dL **no**, this results in neuroglycopenia in hypoglycemic patients

8. A research group investigates the synthesis and release of insulin in beta-cells of pancreas. Which of the following best describes the function of **C-peptide** in this process?

Comment: the C-peptide is **important** for insulin **synthesis** and it is also **measured in blood** by a specific test as it has a longer half-life than insulin.

- A. Needed for release of insulin granules **no** the release is triggered by cellular influx of calcium ions
- B. Is added to insulin to form proinsulin **no** it is cleaved from proinsulin during insulin formation
- C. **Allows optimum formation of all three disulfide bonds** **Yes, correct.** Two of the bonds connect Chain A and Chain B, the third disulfide bond is an intrachain bond in Chain A. The resulting 3-dimensional structure is needed for the insulin receptor recognition.
- D. Increases the half-life of insulin in blood **no** C-peptide is separate from insulin and has a longer half-life than insulin
- E. Links the A- and B-chains of insulin in circulation **no** the C-peptide is released with insulin into the blood and is then degraded.

9. A research group investigates the factors affecting the synthesis and release of **insulin** in pancreatic beta-cells. Which of the following most likely **activates** the release of this hormone?

Comment: it is important to know the regulations of both, insulin and glucagon release and know the incretins that enhance insulin release.

- A. Sympathetic stimulation **no** Glucagon is released during flight and fight situations, insulin is released by parasympathetic stimulation
- B. **Glucagon-like peptide 1 (GLP-1)** **YES, correct GLP-1 is an incretin and activates the release of insulin**
- C. Somatostatin **no**, this hormone inhibits release of both, insulin and glucagon and is involved with fine tuning.
- D. Cortisol **no** this hormone leads to release of epinephrine from the adrenal medulla.
- E. Alanine **no** this amino acid stimulates release of glucagon. Insulin release is stimulated by arginine and leucine.

10. A research group investigates the action of insulin on human skeletal muscle. The volunteers for this study are adolescents aged 14-17 years. All of them are provided a healthy diet containing all dietary essential amino acids. The metabolic changes in skeletal muscle following the meal are observed. Which of the following effects would most likely be observed in these volunteers?

- A. **Increased synthesis of sodium-potassium ATPase. YES, correct it activates the synthesis of sodium-potassium ATPase and with that increases the sodium-dependent uptake of amino acids into muscle for protein synthesis.**

- B. Decreased uptake of amino acids and glucose into muscle cells. **NO** the uptake of amino acids and of glucose from the blood is increased by insulin.
- C. Negative nitrogen balance. **NO**: Leads to a positive nitrogen balance as all dietary essential amino acids are present in the diet of the volunteers and insulin stimulates muscle protein synthesis.
- D. Activation of GLUT-2 transporters for uptake of glucose. **NO**: it activates the GLUT-4 transporters. GLUT-2 is insulin-independent.
- E. Decreased muscle protein synthesis **NO**: Insulin increases muscle protein synthesis

Diabetes mellitus: Practice questions

1. A 45-year-old man comes to the physician for a follow up examination. He has no previous history of diabetes mellitus. His BMI is 35 kg/m² and waist to hip ratio is 1.3. Laboratory studies show plasma glucose levels were 160 mg/dL (Reference range: 70-100 mg/dL) on three previous separate visits. Which of the following is a most likely risk factor for the development of hyperglycemia in this patient?
 - A. Central obesity
 - B. Autoimmune destruction of beta cells
 - C. Adrenal cortical atrophy
 - D. High fat diet
 - E. Low protein diet
2. A 45-year-old woman comes to the physician for a follow up examination. She has a 3-year history of type 2 diabetes mellitus and is on treatment. Her BMI is 32 kg/m² and waist to hip ratio is 1.1. Laboratory studies show plasma glucose is 170 mg/dL (Reference range: 70-100 mg/dL). Which of the following is most likely responsible for the hyperglycemia?
 - A. Increased glucose uptake by the adipocytes
 - B. Increased GLUT-4 in muscle
 - C. Decreased lipoprotein lipase activity
 - D. Increased hepatic glucose output
 - E. Increased insulin sensitivity of adipocytes
3. A 45-year-old man comes to the physician for a follow up examination. He has no family history of diabetes mellitus. His BMI is 38kg/m² and waist to hip ratio is 1.3. Laboratory studies show plasma glucose is 150 mg/dL (Reference range: 70-100 mg/dL) on three previous visits. Which is the most appropriate dietary advice for this patient?
 - A. Increase dietary sucrose intake
 - B. Increase dietary fiber

- C. Increase dietary fat intake to 40% of calories
 - D. Include foods with high glycemic index
 - E. Incorporate a sedentary lifestyle
4. A 54-year-old man comes to the physician for a follow up examination. He has a family history of diabetes mellitus. His BMI is 38kg/m² and waist to hip ratio is 1.2. Laboratory studies show his plasma glucose is 200 mg/dL (Reference range: 70-100 mg/dL) on three separate previous occasions. The physician explains that he is at a high risk for the development of macrovascular complications. Which of the following factors is most likely responsible for these complications?
- A. Decreased HbA1c
 - B. Increased lipoprotein lipase activity
 - C. Increased serum VLDL
 - D. Increased serum urea and creatinine
 - E. Increased glucose loss in urine
5. A 15-year-old-girl is brought to the physician by her parents because of a 2-week history of polyuria, polydipsia and weight loss. Physical findings show mild dehydration. Laboratory tests show fasting plasma glucose is 280 mg/dL and elevated levels of ketone bodies in blood and urine. Which of the following is the most likely diagnosis?
- A. Beta cell tumor secreting insulin
 - B. Cortisol secreting tumor of the adrenal cortex
 - C. Carcinoid syndrome due to tumor of APUD cells
 - D. Pheochromocytoma
 - E. Autoimmune response to a viral infection
 - F. Type 2 diabetes mellitus

Answer key:

- 1. A
- 2. D
- 3. B
- 4. C
- 5. E

Short explanations:

1. A 45-year-old man comes to the physician for a follow up examination. He has no previous history of diabetes mellitus. His BMI is 35 kg/m² and waist to hip ratio is 1.3. Laboratory studies show plasma

glucose levels were 160 mg/dL (Reference range: 70-100 mg/dL) on three previous separate visits. Which of the following is a most likely risk factor for the development of hyperglycemia in this patient?

Comment: The presentation is a typical patient with central obesity (abdominal obesity/ apple shape) which is a risk factor for development of insulin resistance and hyperglycemia. This person has many risk factors for the development of the insulin resistance syndrome (metabolic syndrome/ syndrome X). The target tissues (liver, muscle and adipose tissue) have insulin resistance (not sensitive to insulin).

- A. Central obesity – Correct answer – is a risk factor for development of insulin resistance and type 2 diabetes mellitus**
- B. Autoimmune destruction of beta cells – is seen in type 1 diabetes mellitus
- C. Adrenal cortical atrophy – results in Addison disease characterized by hypoglycemia and hypotension, low serum sodium levels and hyperkalemia
- D. High fat diet
- E. Low protein diet

2. A 45-year-old woman comes to the physician for a follow up examination. She has a 3-year history of type 2 diabetes mellitus and is on treatment. Her BMI is 32 kg/m² and waist to hip ratio is 1.1. Laboratory studies show plasma glucose is 170 mg/dL (Reference range: 70-100 mg/dL). Which of the following is most likely responsible for the hyperglycemia?

Comment: This question is testing the metabolic changes that are responsible for the development of hyperglycemia.

- A. Increased glucose uptake by the adipocytes - The uptake of glucose in adipocytes and muscle is reduced due to less GLUT4.
- B. Increased GLUT-4 in muscle - The uptake of glucose in adipocytes and muscle is reduced due to less GLUT4.
- C. Decreased lipoprotein lipase activity - Lipoprotein lipase is induced by insulin and patients with diabetes mellitus have decreased lipoprotein lipase activity which results in increased VLDL and TAG levels. Decreased lipoprotein lipase activity is NOT responsible for the hyperglycemia.
- D. Increased hepatic glucose output – Correct answer -** Hyperglycemia in type 1 and type 2 diabetes mellitus is due to very low insulin levels (type 1) and low insulin levels and decreased insulin sensitivity of tissues (type 2). As a result, the hepatic glucose output is high and the number of GLUT4 (responsive to insulin) in adipose tissue and muscle is decreased.
- E. Increased insulin sensitivity of adipocytes – The adipocytes, muscle and liver are less sensitive to insulin (insulin resistance) in type 2 diabetics

3. A 45-year-old man comes to the physician for a follow up examination. He has no family history of diabetes mellitus. His BMI is 38kg/m² and waist to hip ratio is 1.3. Laboratory studies show plasma glucose is 150 mg/dL (Reference range: 70-100 mg/dL) on three previous visits. Which is the most appropriate dietary advice for this patient?

Comment: The patient has type 2 diabetes mellitus with obesity. The goal of management of the patient would be to reduce his body weight and reduce central obesity by increasing physical activity and incorporation of regular exercise routines– both of which would improve insulin sensitivity. The other lifestyle modifications are to reduce the dietary consumption of refined sugars (sucrose) and reduce the glycemic index of foods which can be achieved by including dietary fiber – which delays the rise in blood glucose following a meal. Obese persons with type 2 diabetes mellitus (Metabolic syndrome) often have dyslipidemia and increased atherogenic small dense LDL – which increases the risk of atherosclerosis and cardiovascular disease

- A. Increase dietary sucrose intake – Decrease refined sugars and substitute sugar with non-caloric sweeteners
- B. Increase dietary fiber – Correct answer – Increasing dietary fiber content helps to reduce the glycemic index of their diet. It also helps weight management as it provides satiety (a feeling of fullness) but no calories! Also, helps in management of serum cholesterol.**
- C. Increase dietary fat intake to 40% of calories – The ideal goal is to reduce fats to 25-30% of the dietary calories to reduce the risk of atherosclerosis
- D. Include foods with high glycemic index – advise to reduce glycemic index of diet to prevent rapid increase in plasma glucose levels following a meal
- E. Incorporate a sedentary lifestyle – Goal is to reduce his body weight and reduce central obesity by increasing physical activity and incorporation of regular exercise routines– both of which would improve insulin sensitivity

4. A 54-year-old man comes to the physician for a follow up examination. He has a family history of diabetes mellitus. His BMI is 38kg/m² and waist to hip ratio is 1.2. Laboratory studies show his plasma glucose is 200 mg/dL (Reference range: 70-100 mg/dL) on three separate previous occasions. The physician explains that he is at a high risk for the development of macrovascular complications. Which of the following factors is most likely responsible for these complications?

Comment: Obese persons with type 2 diabetes mellitus and obesity (Metabolic syndrome) often have dyslipidemia (Increased VLDL due to decreased lipoprotein lipase activity) and increased atherogenic small dense LDL – which increases the risk of atherosclerosis and cardiovascular disease. Remember, lipoprotein lipase is induced by insulin and requires insulin for optimal activity.

- A. Decreased HbA1c – HbA1c levels are a good indicator of plasma glucose levels over the past 3 months. Most likely this patient may have higher HbA1c levels and higher risk of long-term complications

- B. Increased lipoprotein lipase activity – Type 2 diabetics have low activity of lipoprotein lipase. Insulin induces lipoprotein lipase and insulin resistance results in low activity of the enzyme
- C. Increased serum VLDL – Correct answer** - due to decreased lipoprotein lipase activity; Remember, optimum lipoprotein lipase activity requires insulin. Patients with type 2 diabetes have low insulin sensitivity and reduced activity of lipoprotein lipase
- D. Increased serum urea and creatinine – This is seen in patients with renal failure
- E. Increased glucose loss in urine – Plasma glucose levels are higher than the renal threshold for glucose. Glycosuria increases the risk of urinary and bladder infections and dehydration.

5. A 15-year-old-girl is brought to the physician by her parents because of a 2-week history of polyuria, polydipsia and weight loss. Physical findings show mild dehydration. Laboratory tests show fasting plasma glucose is 280 mg/dL and elevated levels of ketone bodies in blood and urine. Which of the following is the most likely diagnosis?

Comment: This is a classical presentation of type 1 diabetes mellitus – polyuria, polydipsia, hyperglycemia with ketosis. The beta cell destruction is triggered by an autoimmune response which follows a viral infection. Ketosis is more common in type 1 diabetics than in type 2 diabetics.

- A. Beta cell tumor secreting insulin - Beta cell tumor (insulinoma) causes hypoglycemia
- B. Cortisol secreting tumor of the adrenal cortex - Patients with Cushing syndrome have hyperglycemia (secondary diabetes) accompanied by changes in fat distribution, skin changes and other features.
- C. Carcinoid syndrome due to tumor of APUD cells - Patients with Carcinoid syndrome have episodes of diarrhea and flushing (no hyperglycemia).
- D. Pheochromocytoma - Patients with pheochromocytoma usually have episodes of hypertension and hyperglycemia.
- E. Autoimmune response to a viral infection – Correct answer** - The beta cell destruction is triggered by an autoimmune response which follows a viral infection.
- F. Type 2 diabetes mellitus - Ketosis is not very common in type 2 diabetes mellitus.